S&DS 365 / 665

Intermediate Machine Learning

Course Overview

Wednesday, September 2

Yale

Welcome!

- Overview of course
- Topics
- Syllabus and logistics
- Start on sparse regression (if time)

Course objectives

Gain a solid understanding of concepts and methods of modern machine learning

- Become comfortable with the core ideas
- Gain an understanding of:
  - How and why methods work (and often don't!)
  - Which models are appropriate for a given problem
  - How to adapt and extend methods when needed
- Get close to the research frontier

What does “Intermediate” imply?

- A second course in machine learning
- Assume familiar with things like PCA, bias/variance, maximum likelihood, basics of neural nets
- Have experimented with basic ML methods on some data sets
- Previous exposure to Python
- More on this later...

What's the difference between 365 and 265?[†]

- Little overlap in topics
- 365 (IML) is more technical/mathematical than 265 (iML)
- We'll do some theory (but not for theory's sake!)
- The courses have similar organization (lectures, demos, assignments...)
- IML is required for S&DS majors (BS); iML aims to be accessible to broad cross-section of Yale students
- More on differences later...

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† 100

Course materials

Materials posted to `http://interml.ydata123.org`; sometimes to Canvas

iML page: `https://ydata123.org/sp26/introml/calendar.html`

Please use Ed Discussion for any questions about lectures, homework, etc.

Please use email `sds365@yale.edu` for all course questions

What is Machine Learning?

The study of algorithms and statistical models to develop computer programs that improve with experience.

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The study of algorithms and statistical models to develop *computer programs that create computer programs* that improve with experience.

What is Machine Learning?

Machine Learning is closely aligned with Statistics, but with a focus on computation, scalability, prediction, representation, and complex problems

- Object recognition and scene classification
- Translation from one language to another
- Autonomous driving...

Subproblems of these and other complex problems are concrete, statistical estimation and inference problems that can be studied in isolation.

AI vs. ML

Machine learning focuses on making predictions and inferences from data.

AI combines machine learning components into a larger system that includes a decision making component.

An AI system exhibits a behavior, resulting from the collective decisions that are made.

AI vs. ML: An analogy

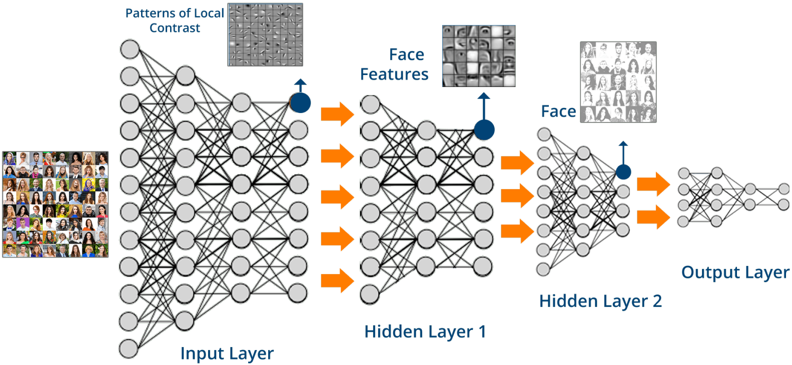
AI $\approx$ nuclear power. Tied up with issues of engineering at scale, economics, jobs, politics and legal infrastructure.

ML $\approx$ nuclear physics. You can't build a nuclear power plant without a handle on the physics of nuclear fission, isotope separation and uranium enrichment.

Machine learning frameworks

- Supervised learning
- Unsupervised (and self-supervised) learning
- Reinforcement learning
- Representation learning

Deep learning is a type of machine learning



- Heuristics motivated from simplified view of the brain
- A particular form of nonlinear classification/regression/density estimation

This course

- Part 1: Supervised learning
- Part 2: Unsupervised learning
- Part 3: Reinforcement learning
- Part 4: Sequence learning

Two types of intelligence

1. Sensory ("Neocortical")— acquire semantic and procedural knowledge
   - Requires extensive data and training
   - Slow to learn, fast to apply
   - Well captured by modern deep learning

Two types of intelligence

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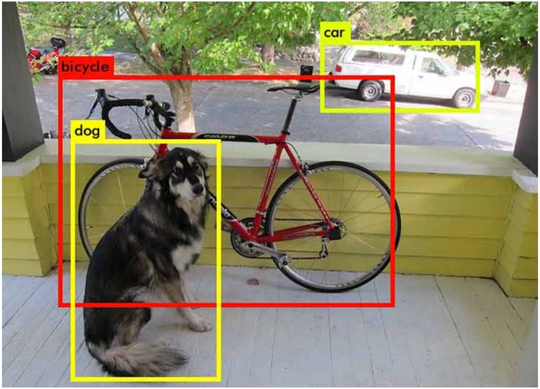
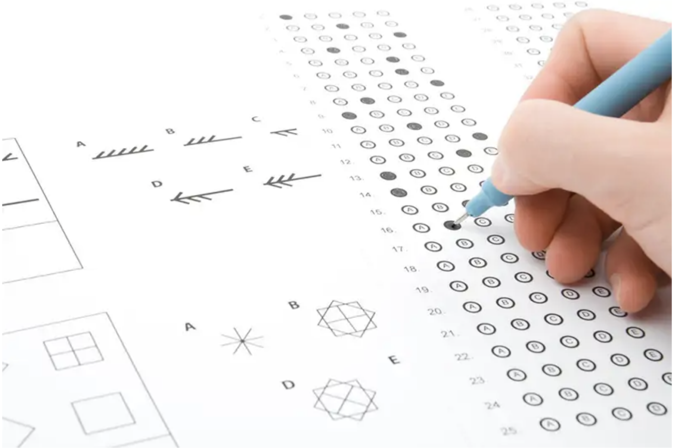


Two types of intelligence

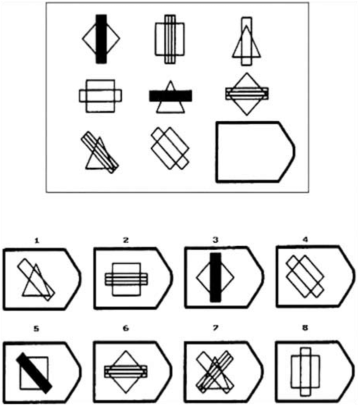
2. Relational ("Prefrontal")— identify novel associations and relations
   - Fast to learn, slow to apply
   - Symbolic processing and abstraction
   - Little explicit training data

Two types of intelligence

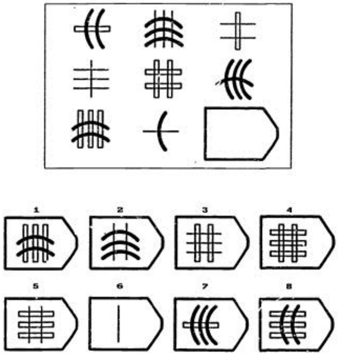


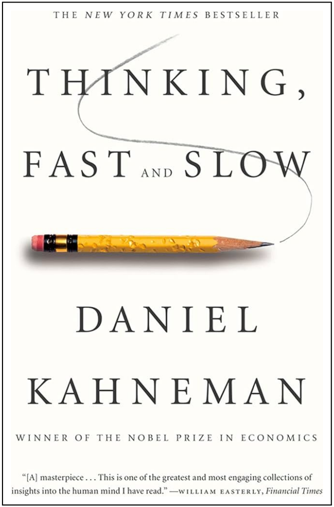
AI struggles with IQ tests
Panther Media GmbH / Alamy Stock Photo

Two types of intelligence



Two types of intelligence





- Sensory (Neocortical): slow to learn, fast to apply
- Relational (Prefrontal): fast to learn, slow to apply

Two types of intelligence

1. Sensory (“Neocortical”)— efficient function approximation
2. Relational (“Prefrontal”)— rapid abstraction, association and generalization

Can both types be supported in a single architecture?

Relations are essential to reasoning

"Two shapes are in a room. The shape next to the cube is blue. The shape next to the sphere is red. Which is red, the cube or the sphere?"

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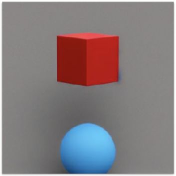

Relations are essential to reasoning

Based on the given information, it can be inferred that the sphere is red. The statement specifies that the shape next to the sphere is red, implying that the sphere itself is red. The color of the cube is not mentioned, so we cannot determine its color from the given information.

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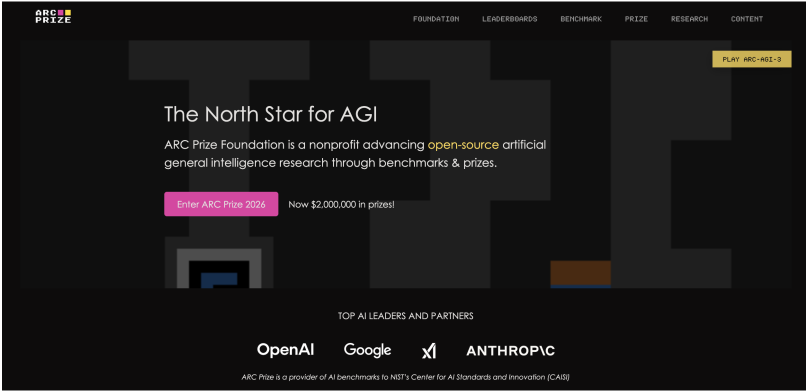
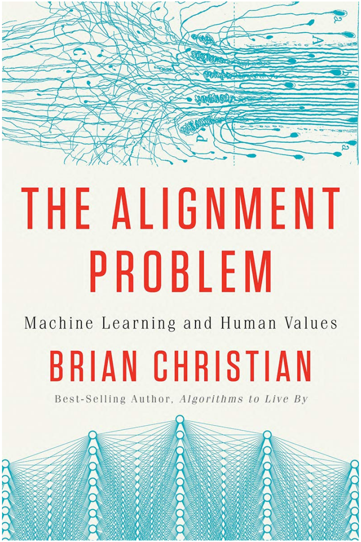
Abstraction and Reasoning Prize

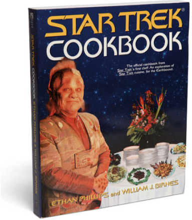


Shortcomings are masked

- Recent innovations in AI bots hide these deficiencies
- Systems are trained to convince us
- Can lead to over-confidence and bogus reasoning
- After this course you'll have an understanding of core components of AI systems

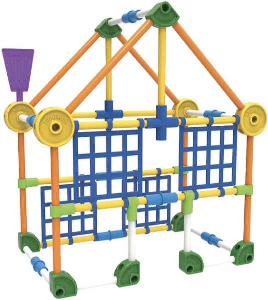

THE ALIGNMENT PROBLEM
Machine Learning and Human Values
BRIAN CHRISTIAN
Best-Selling Author, Algorithms to Live By

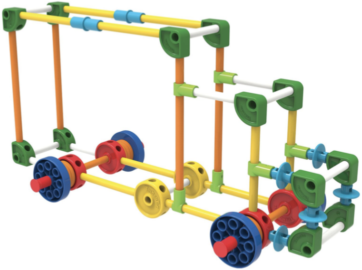
Machine learning *modus operandi* **(1/2)**



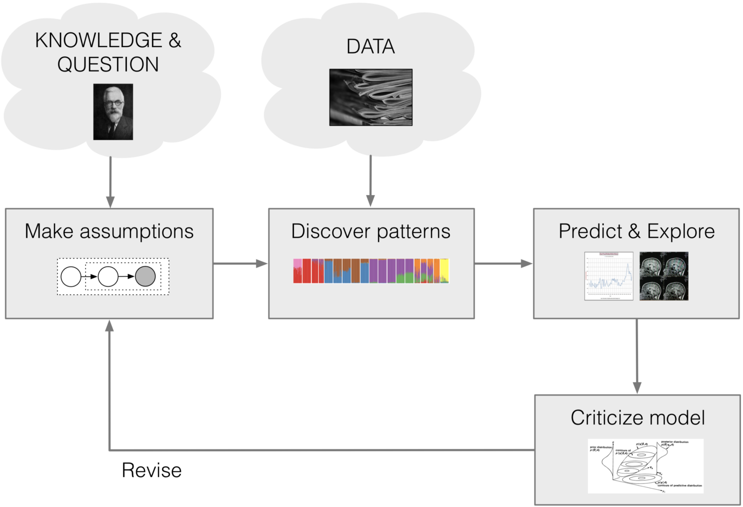
Increasingly powerful computing platforms (Keras, Tensorflow, Pytorch) implementing a few standard architectures.

Machine learning *modus operandi* (2/2)





More challenging to design and implement. Computation typically not "off the shelf"



Ack: Dave Blei

Logistics

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Syllabus

Intermediate Machine Learning is a second course in machine learning at the advanced undergraduate or beginning graduate level. The course assumes familiarity with the basic ideas and techniques in machine learning, for example as covered in S&DS 265. The course treats methods together with mathematical frameworks that provide intuition and justifications for how and when the methods work. Assignments give students hands-on experience with machine learning techniques, to build the skills needed to adapt approaches to new problems.

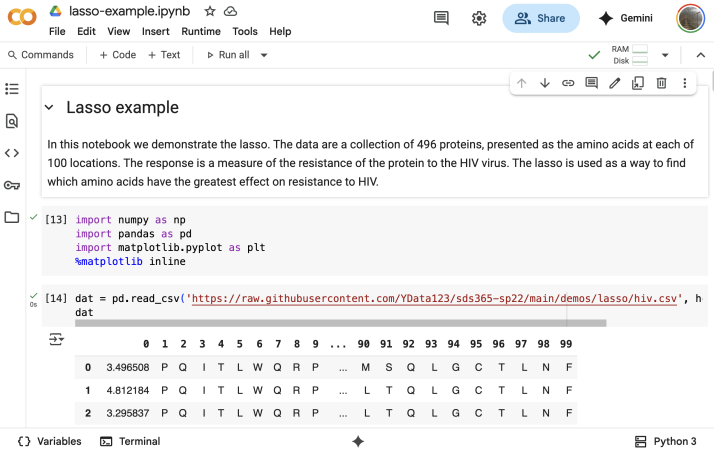
Syllabus

Topics include nonparametric regression and classification, kernel methods, risk bounds, nonparametric Bayesian approaches, graphical models, attention and language models, generative models, sparsity and manifolds, and reinforcement learning. Programming is central to the course, and is based on the Python programming language and Jupyter notebooks.

Prerequisites

- Background in probability and statistics, at the level of S&DS 242 (Theory of Statistics)
- Familiarity with the core ideas from linear algebra, for example through Math 222 (Linear Algebra with Applications)
- Computational skills at the level of S&DS 265 (Introductory Machine Learning) or CPSC 200 (Introduction to Information Systems)
- Previous familiarity with Python is recommended

Running on Google Colab (preferred)



Running on Google Colab (preferred)

Installing Jupyter locally (not supported)

- See installation guide on course Canvas site: `Files > Getting started`
- Use Python 3.x version

Assessment

- Assignments (35%)
- Mid-semester exam (20%)
- End-of-semester exam (20%)
- Project (15%)
- Quizzes (10%)

Assignments

- Five assignments total
- Roughly every 2 weeks
- Due at 11:59pm on a Thursday
- Late assignments not accepted
- Submitted using Gradescope
- Mix of concepts, problem solving and data analysis
- Prepared using Python notebooks

Policy on Collaboration

1. Collaboration on homework assignments with fellow students is encouraged.
2. However, such collaboration should be clearly acknowledged, by listing the names of the students with whom you have had any discussions concerning the problem.
3. You may not share written work or code—after discussing a problem with others, the solution must be written by yourself.
4. Treat an AI assistant as a fellow student. That is, use interactively to help develop solutions, but do not copy/paste code and written solutions from the AI assistant. Acknowledge all collaborations with an AI assistant.

Quizzes

- Five quizzes total
- Taken online (Canvas)
- Short, 10-20 minutes
- Assess understanding of essentials

http://interml.ydata123.org

Week	Dates	Topics	Demos & Tutorials	Lecture Slides	Readings & Notes	Assignments & Exams
1	Sep 2, Sep 4	Sparse regression	CO Python elements CO Pandas and regression CO Lasso example	Wed: Course overview Fri: Sparse regression	Google Colab Basics PML Section 11.4 Notes on linear regression	
2	Sep 9	Smoothing and kernels	CO Smoothing example CO Using different kernels	Wed: Lasso (continued) and smoothing	PML Sections 16.3, 17.1 Notes on computing the lasso	Quiz 1
3	Sep 14, 16	Density estimation and Mercer kernels	CO Density estimation demo CO Mercer kernels (1/3) CO Mercer kernels (2/3) CO Mercer kernels (3/3)	Mon: Smoothing and density estimation Wed: Mercer kernels	Risk bounds for local smoothing Notes on Mercer kernels	CO Assn 1
4	Sep 21, 23	Neural networks and overparameterized models	CO np-complete example (1/2) CO np-complete example (2/2) TensorFlow playground	Mon: Neural networks Wed: Double descent	PML Sections 13.1, 13.2 Notes on backpropagation Notes on double descent	Quiz 2

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5	Sep 28, 30	Gaussian processes and approximate inference	Bayes CO Gaussian processes CO Gibbs sampling for image denoising	processes Wed: Recap of GPs Introduction to approximate inference	Notes on Bayesian inference Notes on nonparametric Bayes Notes on simulation	Assn 1 in CO Assn 2 out
6	Oct 5, 7	Variational inference	CO Variational autoencoders	Mon: Variational inference Wed: VAEs	PML Section 20.3 Notes on variational inference	Quiz 3
7	Oct 12, 14	Graphs and structure learning	CO Graphical lasso demo	Mon: Sparsity and graphs Wed: Discrete data and graph neural nets	Notes on graphs and structure learning Graph neural networks PML Section 23.4	Assn 2 in CO Assn 3 out
8	Oct 19	Exam 1			Practice exams	

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9	Oct 26, 28	Deep reinforcement learning	CO Q-learning demo CO DQN demo	Mon: Reinforcement learning Wed: Deep reinforcement learning	Sutton and Barto, Section 6.5	Assn 3 in CO Assn 4 out
10	Nov 2, 4	Sequential models	CO vanilla RNN CO Fakespeare RNN	Mon: HMMs and RNNs Wed: RNNs, GRUs, LSTMs, and all that	TensorFlow: Text generation Notes on HMMs and Kalman filters PML Chapter 15	Quiz 4 Project proposals due
11	Nov 9, 11	Sequence-to-sequence models and Transformers	CO Transformer demo Minimal LLM decoder	Mon: Sequence models and attention Wed: Transformers, LLM scaling PML Sections 15.4, 15.5		Assn 4 in CO Assn 5 out
12	Nov 16, 18	Diffusion models		Mon: Mathematics of diffusion Wed: Generative modeling		Quiz 5

Exams

- Exam #1: Monday, October 19, in class
- Exam #2: Wednesday, December 9, in class

No rescheduled exams to accommodate early travel plans

Projects

- Explore concepts from course in an area of interest to you
- Can work together in teams of up to three students
- AI tools are allowed (and encouraged!)
- Project proposals due mid-semester (TBD)
- Assessment:
  - Problem statement
  - Data set
  - Key machine learning method(s)
  - Results
  - Statement on AI use

Auditing

- Auditors are welcome!
- Full access to Canvas
- Just expected to regularly attend class

Questions on logistics?

Topics covered

- Part 1: Supervised learning
- Part 2: Unsupervised learning
- Part 3: Reinforcement learning
- Part 4: Sequence learning

Topics covered

- *Part 1: Supervised learning*
  - Sparse regression
  - Smoothing and kernels
  - Risk bounds and generalization error
  - Some theory for deep learning
- Part 2: Unsupervised learning
- Part 3: Reinforcement learning
- Part 4: Sequence learning

Topics covered

- Part 1: Supervised learning
- *Part 2: Unsupervised learning*
  - Nonparametric Bayes
  - Approximate inference
  - Approaches to generative models
  - Structure learning
  - Diffusion models
- Part 3: Reinforcement learning
- Part 4: Sequence learning

Topics covered

- Part 1: Supervised learning
- Part 2: Unsupervised learning
- *Part 3: Reinforcement learning*
  - Deep Q-Learning
  - Policy gradient methods
- Part 4: Sequence learning

Topics covered

- Part 1: Supervised learning
- Part 2: Unsupervised learning
- Part 3: Reinforcement learning
- *Part 4: Sequence learning*
  - Classical techniques (Kalman filters, HMMs)
  - Recurrent neural networks
  - Attention and language models
  - Transformers

References

- "Probabilistic Machine Learning: An Introduction," by K. Murphy, MIT Press, `https://probml.github.io/pml-book/book1.html`
- "Probabilistic Machine Learning: Advanced Topics," by K. Murphy, MIT Press, `https://probml.github.io/pml-book/book2.html`
- "The Elements of Statistical Learning: Data Mining, Inference, and Prediction," by T. Hastie, R. Tibshirani, and J. Friedman,

  `http://www-stat.stanford.edu/~tibs/ElemStatLearn/`

Questions?

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Remember: Class on Friday!